

Ferroelectric HfO₂ Memory — Ontology Specification

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- Every class, property and individual carries a source tag in the form **S1{I A}** = source S1, Section I A.
- $\Rightarrow$ **UNCERTAIN** marks something that is contested between sources, implied rather than stated, out of context, or otherwise not safe to assert.
- $\times$ **INFERRED** marks modelling structure I introduced that no source states. These are design decisions, not findings.
- Anything without a tag is ontology scaffolding, not a claim about the world.

1 Classes

Indentation = `rdfs:subClassOf`. Curly braces list verified individuals.

```
Thing
|
+-- Material
|
|   +-- Substrate
|   |   { Si_wafer, Si_001, Si_p_doped_heavy, Si_3inch_10ohmcm,
|   |   Si_8inch_p_type, SiO2_on_Si_PECVD }          S1{II}, S9, S11, S23, S24, S27
|   |   [UNCERTAIN] Epitaxial substrates (YSZ, LSMO, ITO, scandates) appear only in
|   |   S1{II C} as PLD research substrates, not production  S1{II C}
|   |
|   +-- Ferroelectric_Material
|   |   +-- Undoped_HfO2          { HfO2 }          S1{I B}
|   |   +-- Doped_HfO2
|   |   |   { Si_HfO2, Y_HfO2, Sr_HfO2, La_HfO2, Gd_HfO2, Al_HfO2,
|   |   |   N_HfO2, Fe_HfO2, Sc_HfO2, Ge_HfO2 }      S1{I B, II B 1}
|   |   +-- Hafnium_Zirconate    { HZO }          S1{I}
|   |   |   [UNCERTAIN] "HZO" is a composition range, not a fixed formula.
|   |   |   S24 measured Hf:Zr ~ 0.8, i.e. Hf0.45Zr0.55O2, and still
|   |   |   calls it HZO. Use hasHfZrRatio, do not infer 50:50.  S24{2}
|   |   +-- Doped_Hafnium_Zirconate { La_HZO, Al_codoped_HZO } S27, S6
|   |   +-- Hafnium_Aluminate     { HAO } Al:Hf = 1:30  S9{p.2}
|   |
|   +-- Electrode_Material
|   |   { TiN, TaN, Ru, W, Pt, IrO2, Al }          S1{II A 2}, S12a, S24
|   |   [UNCERTAIN] In the polycrystalline MFM literature Ti/Pt is a contact pad,
|   |   but in PLD and CSD literature Pt is a functional electrode S1{II}
|   |
|   +-- Dielectric_Layer_Material
|   |   +-- Interfacial_Layer_Material { SiO2, SiOx, SiON }      S9, S10, S11
|   |   +-- Insertion_Layer_Material   { Al2O3 }              S1{II A 3}, S11
|   |   +-- Seed_Layer_Material        { HfO2_seed, ZrO2_seed } S1{II A 3}
|   |   +-- Interface_Reaction_Product { TiO2, HfN }          S1{II B 2}
|   |   |   TiO2 forms deliberately (pre-deposition O2) or parasitically
|   |   |   (excess O3). HfN is the defect this prevents.
|   |
|   +-- Channel_Material
```

		{ Si_bulk, polySi, IGZO }	S5, S6
		[UNCERTAIN] ITO and IWO named by S6 as a class of oxide semiconductors	
		under investigation; no device data verified	S6
		+-- Chemical_Precursor	
		+-- Hf_Precursor { HfCl4, TDMA_Hf, TEMA_Hf, CpHf_NMe2_3 }	S1{II A 1}
		+-- Zr_Precursor { TDMA_Zr, TEMA_Zr, CpZr_NMe2_3 }	S1{II A 1}
		+-- Dopant_Precursor	S1{Table I}
		[UNCERTAIN] Individual dopant precursors are listed in S1 Table I but	
		none was verified in a device flow. Treat as catalogue.	
		+-- Electrode_Precursor { TiCl4, NH3, Ru_EtCp_2 }	S23{2}, S24{2}
		+-- Oxidant { H2O, O3, O2_gas, O2_plasma, H2O2 }	S1{II A 2}, S23{2}
		+-- Process_Gas	
		{ N2, Ar, O2, NH3, forming_gas_5H2_95N2, SF6 }	S1, S9, S10, S24
		+-- Sputter_Target	
		{ HfO2_target, ZrO2_target, Hf_metal, Y2O3_target, HZO_target }	S1{II B, Table III}
		+-- Wet_Chemical { SC1, dilute_HF }	S1{II}, S10, S11
		+-- Dopant { Si4, Y3, Sr2, La3, Gd3, Al3, N3, Zr4,	
		Fe3, Sc, Ge, Ce, Nd, Sm, Er, Yb }	S1{I B, II B 1, II D}
		+-- Phase_State	
		+-- Amorphous	S1{II A 1}
		+-- Crystalline_Phase	
		+-- Orthorhombic_Pca21 ferroelectric	S1{I A}
		+-- Orthorhombic_Pbca non-ferroelectric	S1{II E}
		+-- Tetragonal_P42nmc non-ferroelectric	S1{I A}
		+-- Monoclinic_P21c non-ferroelectric, bulk-stable	S1{I A}
		+-- Cubic_Fm3m	S1{I A}
		+-- Defect	
		+-- Point_Defect	
		+-- Oxygen_Vacancy stabilises o- and t-phase	S1{I B, II A 2}
		+-- Oxygen_Interstitial removes barrier to m-phase	S1{II A 2}
		+-- Metal_Vacancy Hf, Zr -- origin of deep traps	S10{III D}
		+-- Electronic_Trap { Et1, Et2 }	S10{III D}
		+-- Layer_Stack	
		+-- MFM metal-ferroelectric-metal	S1{II}
		+-- MFIS metal-ferroelectric-insulator-semiconductor	S9, S10, S11
		+-- MIFIS metal-insulator-ferroelectric-insulator-semiconductor	S11{II}
		+-- MFMIS metal-ferroelectric-metal-insulator-semiconductor	S5, S6
		+-- Nanolaminate	S1{II A 3}
		+-- Bilayer	S1{II A 3}
		+-- Trilayer	S1{II A 3}
		+-- Heterogeneous_Co_Doped_Stack e.g. IL/HAO/HAO/HZO	S9{p.2}
		+-- Memory_Cell_Architecture	
		+-- Capacitor_Based_Cell	
		+-- FeCAP_1C	S12a{Fig. 1b}
		+-- FeRAM_1T1C_Charge_Sensed destructive read	S5
		+-- FeRAM_1T1C_Capacitance_Sensed non-destructive, <2x ratio	S5
		+-- Transistor_Based_Cell	
		+-- FeFET_1T	S5
		+-- FeMFET_1T1C MFMIS	S5
		+-- Hybrid_Cell	
		+-- FeRAM_2T1C	S5
		+-- FeRAM_2TnC quasi-non-destructive readout	S5
		+-- FeRAM_1TnC	S5

```

+-- FTJ
| [UNCERTAIN] NAMED ONLY. Listed as an application of ferroelectric HfO2
| by S1{I} and S27{p.1134}. No structure, sensing mechanism or
| performance data verified. Do not populate.
|
+-- Array_Architecture
| +-- VC_FeNAND      vertical channel      S5
| +-- HC_FeNAND      horizontal channel    S5
| +-- Fe_AND         S5
| +-- Stacking_Strategy
|   +-- Parallel_Stacking lithography count ~independent of layers S5
|   +-- Sequential_Stacking lithography scales ~linearly with layers S5
|
+-- Process_Flow
| { FlowA_LabMFM, FlowB1_GateLastFeFET, FlowB2_GateLastFeFET_MIFIS,
|   FlowC_BEOL_FeRAM, FlowD_MFIS_Capacitor, FlowE_FullALD_AutoAnneal,
|   FlowF_MFM, FlowG_LaHZO_MIM } S4
|
+-- Unit_Process
| +-- Substrate_Preparation
| | +-- Wet_Clean      { RCA_SC1, dHF_clean } S1, S10, S11
| | +-- Native_Oxide_Removal S9{p.2}
| | +-- Pre_Deposition_Oxygen_Flow forms self-limiting TiO2 S1{II B 2}
| +-- Interfacial_Layer_Formation
| | +-- Ozone_Oxidation      O3, 300 C, in ALD chamber S10{II A}, S11{II}
| | +-- Chemical_Oxidation_Plus_Nitridation RTP in NH3, then O2/N2 S9{p.2}
| +-- Deposition
| | +-- ALD
| | | +-- Thermal_ALD S24{2}
| | | +-- Plasma_Enhanced_ALD S23{2}
| | | +-- Radical_Enhanced_ALD Ru from Ru(EtCp)2 + O* S24{2}
| | +-- PVD
| | | +-- Sputter_Deposition { DC, RF, magnetron, reactive } S1{II B}
| | | +-- Evaporation S1{II}
| | +-- PLD S1{II C}
| | +-- CSD { spin_coat, dip_coat } S1{II D}
| | +-- PECVD SiO2 insulation S24{2}
| +-- Thermal_Process
| | +-- Crystallisation_Anneal RTA/RTP S1{II}, S9-S12a, S23
| | +-- Post_Metallisation_Anneal furnace, 1 h S27{p.1134}
| | +-- Dopant_Activation_Anneal 1050 C / 5 s S10, S11
| | +-- Forming_Gas_Anneal 450 C, 5% H2/95% N2 S10, S11
| | +-- Retention_Bake 85 C diagnostic retention S24{3.3}
| |   105 C fatigue recovery test S24{3.2}
| +-- Implantation channel doping; S/D formation S10, S11
| +-- Lithography { photolithography, hard_mask } S1{II}, S24{2}
| +-- Etch
| | +-- Wet_Etch { SC1_TiN_removal } S1{II}
| | +-- Plasma_Etch { SF6_Ar_TiN, O2_Ar_Ru } S24{2}
| +-- BEOLProcess
| | +-- Isolation_Deposition S12a{Fig. 1a}
| | +-- Via_Patterning_And_Etch S12a{Fig. 1a}
| | +-- Metal_Deposition_And_Patterning S12a{Fig. 1a}
| +-- Contact_Formation { TiN_Al, Al_pads } S10, S24
| +-- Electrical_Conditioning
| | +-- Wake_Up_Cycling 10~5 cycles, 3 us, 3 V S24{3.2}
| | [UNCERTAIN] Verified as an experimental preparation step, not as a
| | production process step. v1 claimed the latter on an
| | unobtainable source. Model as a conditioning step used
| | before characterisation; do not assert it is in any
| | manufacturing flow.
|
+-- Equipment

```

```

| +-- Deposition_Tool
| | { ALD_reactor, PEALD_reactor, REALD_reactor,
| |   DC_magnetron_sputter, RF_magnetron_sputter, PLD_chamber,
| |   PECVD_tool, spin_coater }
| |   Verified instance: Picosun R200adv
| |   S1, S23, S24
| |   S24{2}
| +-- Thermal_Tool
| | { RTP_system, furnace }
| |   S1{II}, S9, S27
| |   [INFERRED] v1 listed a UV nanosecond laser anneal tool from an
| |   unverified source. Removed.
| +-- Lithography_Tool
| |   S1{III}, S24{2}
| |   [UNCERTAIN] No verified source names a specific exposure tool. v1's
| |   "DUV scanner vs mask aligner" distinction is unsupported.
| +-- Etch_Tool
| |   plasma etcher
| |   S24{2}
| +-- Implant_Tool
| |   ion implanter
| |   S10, S11
| +-- Metrology_Tool
| | { GIXRD_diffractionmeter, TEM, HRTEM, PED_ASTAR_system,
| |   ToF_SIMS, XPS, EDS }
| |   S1, S9, S24, S27
| |   Verified instances: ARL X'TRA (GIXRD) S24{2};
| |   Thermo Fisher Titan3 G2 60-300 + Nanomegas DIGISTAR P2010
| |   with Topspin/ASTAR + DensSolutions heating-biasing holder
| |   S27{p.1134}
| +-- Electrical_Test_Tool
| | { probe_station, ferroelectric_tester, parameter_analyzer,
| |   pulse_generator }
| |   S10, S24, S23
| |   Verified instances: Keysight/Agilent B1500A S10{II A}, S11{III},
| |   S24{2}; Radiant Precision LC S10{II A};
| |   Summit 11000B-M probe station S24{2};
| |   F3000, Zhejiang Liryder S23{2}
|
| +-- Capability
| | { Conformality, SubNanometreThicknessControl, LineOfSight,
| |   SelfLimitingGrowth, HighDepositionRate, LowCarbonContamination,
| |   EtchSelectivityToHfO2, EpitaxialGrowth }
| |   S1{II A, II B, III}, S24{1}
|
| +-- Measurement_Method
| | +-- PE_Hysteresis
| |   S10, S24
| | +-- PUND
| |   S9, S23, S24
| | +-- NDPU
| |   S23
| | +-- Dynamic_IswE
| |   S24{2}
| | +-- Small_Signal_CV
| |   S24{2}
| | +-- QSCV
| |   S10{II C}
| | +-- Endurance_Cycling
| |   S9-S12a, S24
| | +-- Retention_Measurement
| |   SS / NSS / OS states
| |   S24{3.3}
| | +-- IdVg_Transfer
| |   S10, S11
| | +-- Permittivity_From_Displacement_Current
| |   S23{3.2}
| | +-- GIXRD_Phase_Analysis
| |   S9, S24
| | +-- PED_Orientation_Mapping
| |   S27
| | +-- ToF_SIMS_Depth_Profiling
| |   S9
|
| +-- Measurement
| |   [INFERRED] REIFIED n-ary class. See S3.1
|
| +-- Measured_Property
| | +-- Polarisation_Property
| | | { remanent_polarisation_Pr, double_remanent_2Pr,
| | |   switched_polarisation_Psw, saturated_polarisation_Ps }
| | +-- Field_Property
| | | { coercive_field_Ec, coercive_voltage_Vc, bias_field_Ebias,
| | |   depolarisation_field_Ed, interfacial_field_EI,
| | |   built_in_field_Ebuilt_in, breakdown_field }
| | +-- Device_Property
| | | { memory_window_MW, threshold_voltage_Vth, sense_margin,
| | |   bitline_voltage_VBL, switching_speed, on_current_Ion }
| | +-- Dielectric_Property
| | | { relative_permittivity_k, leakage_current_density }

```

```

| +-- Reliability_Property
| | { endurance_cycles, retention_time, imprint_shift,
| |   wakeup_magnitude, trapped_charge_density }
| +-- Structural_Property
| | { phase_fraction, grain_size, grain_aspect, film_thickness,
| |   carbon_impurity, oxygen_vacancy_concentration,
| |   growth_per_cycle, film_density, crystallographic_texture }
|
+-- Degradation_Phenomenon
+-- Wake_Up
| +-- First_Stage_Wake_Up S24{3.2}
| +-- Second_Stage_Wake_Up activates >343 K at 10~7-10~8 cycles S24{3.2}
+-- Wake_Up_Relaxation Vo driven back during idle at 0 V S23{4}
+-- Fatigue S24{3.2}
+-- Imprint S23, S24
| [UNCERTAIN] "Fluid imprint" reported by S6 as a distinct rapid mechanism
| under repeated non-destructive read. Trade-press only. S6
+-- Retention_Loss
| +-- Relaxation ruled out in S24's devices S24{3.3}
| +-- Depolarisation_Loss S24{3.3}
| +-- Imprint_Driven_Loss S24{3.3}
+-- Charge_Trapping S5, S10, S11
+-- Dielectric_Breakdown S24{3.2}
+-- Pass_Disturb 3D NAND strings S5
+-- Read_Disturb accumulative switching in QNRO S5
| [UNCERTAIN] "Localised phase variation -> Vth variation" reported by S6
| only (trade press). Not modelled as a class.

```

2 Object properties

2.1 Process ordering

Property	Domain	Range	Characteristics	Notes
belongsToFlow	Unit_Process	Process_Flow	—	Required on every Unit_Process individual. Without it, precedes produces contradictions between Flow C and the rest
directlyPrecedes	Unit_Process	Unit_Process	asymmetric, ir-reflexive	Immediate ordering within one flow
precedes	Unit_Process	Unit_Process	transitive, asymmetric, irreflexive	Transitive closure of directlyPrecedes , scoped by flow
hasStepIndex	—	—	—	Alternative to precedes if you prefer integer ordering
flowProducesArchitecture	Process_Flow	Memory_Cell_Architecture ∪ Layer_Stack	—	Flow C → FeRAM_1T1C; Flow E → MFM; Flow B1 → MFIS
flowTargetsNode	Process_Flow	Technology_Node	—	Flow C → 130 nm S12a{¶4}

Verified ordering facts:

- `directlyPrecedes(Top_Electrode_Deposition, Crystallisation_Anneal)` holds in Flows A, C, D, F, G and B1/B2. `s1`, `s9`, `s10`, `s11`, `s12a`, `s23`, `s27`
- `precedes(Crystallisation_Anneal, Patterning)` holds in **A, B1, B2, D, F, G**. (Flow E has no `Crystallisation_Anneal` individual; instead, `E5_TiN_TE_ALD` — which `transformsPhase` — precedes `E6_Patterning`.)
- `precedes(Patterning, Crystallisation_Anneal)` holds in **Flow C alone**. `s12a`{Fig. 1a}
- `precedes(Dopant_Activation_Anneal, Interfacial_Layer_Formation)` holds in the gate-last flows B1, B2. `s10`, `s11`
- Flow E has **no** `Crystallisation_Anneal` individual at all; crystallisation occurs during the ALD TiN top electrode deposition (E5), which is a `Unit_Process` with `transformsPhase = true`. `s24`{2}

2.2 Process composition

Property	Domain	Range	Notes
<code>hasInput</code>	<code>Unit_Process</code>	<code>Material</code> $\cup$ <code>Artifact</code>	Many-to-many
<code>hasOutput</code>	<code>Unit_Process</code>	<code>Material</code> $\cup$ <code>Artifact</code>	
<code>usesEquipment</code>	<code>Unit_Process</code>	<code>Equipment</code>	Many-to-many. One ALD reactor serves electrode and ferro-electric deposition <code>s24</code> {2} (<code>S24</code> has no interfacial layer)
<code>usesPrecursor</code>	<code>Deposition</code>	<code>Chemical_Precursor</code>	
<code>usesOxidant</code>	<code>Deposition</code>	<code>Oxidant</code>	
<code>usesProcessGas</code>	<code>Unit_Process</code>	<code>Process_Gas</code>	Covers carrier, sputter, anneal ambient and etch gases
<code>usesTarget</code>	<code>Sputter_Deposition</code>	<code>Sputter_Target</code>	
<code>depositsMaterial</code>	<code>Deposition</code>	<code>Material</code>	
<code>etchesMaterial</code>	<code>Etch</code>	<code>Material</code>	<code>SF6_Ar_TiN etchesMaterial TiN</code> <code>s24</code> {2}
<code>hasAmbient</code>	<code>Thermal_Process</code>	<code>Process_Gas</code>	<code>N2</code> recommended over <code>O2</code> <code>s1</code> {III B 2}

2.3 Structure and composition

Property	Domain	Range	Notes
<code>hasLayer</code>	<code>Layer_Stack</code>	<code>Material</code>	Ordered. Use <code>hasLayerAtPosition</code> or an RDF list; bare <code>hasLayer</code> loses the stacking order that <code>S9</code> 's whole result depends on
<code>hasLayerAtPosition</code>	<code>Layer_Stack</code>	<code>Material</code>	Qualified by <code>layerIndex</code>
<code>hasElectrode</code>	<code>Layer_Stack</code>	<code>Electrode_Material</code>	Sub-properties <code>hasTopElectrode</code> , <code>hasBottomElectrode</code>
<code>hasInterfacialLayer</code>	<code>Layer_Stack</code>	<code>Interfacial_Layer_Material</code>	
<code>hasFerroelectricLayer</code>	<code>Layer_Stack</code>	<code>Ferroelectric_Material</code>	
<code>hasChannel</code>	<code>Memory_Cell_Architecture</code>	<code>Channel_Material</code>	
<code>usesStack</code>	<code>Memory_Cell_Architecture</code>	<code>Layer_Stack</code>	<code>FeFET_1T</code> $\rightarrow$ <code>MFIS</code> ; <code>FeMFET_1T1C</code> $\rightarrow$ <code>MFMIS</code> <code>s5</code>
<code>hasDopant</code>	<code>Ferroelectric_Material</code>	<code>Dopant</code>	
<code>hasPhase</code>	<code>Material</code> $\cup$ <code>Layer_Stack</code>	<code>Phase_State</code>	$\Rightarrow$ UNCERTAIN Films are phase <i>mixtures</i> .
<code>arrangedAs</code>	<code>Memory_Cell_Architecture</code>	<code>Array_Architecture</code>	

Property	Domain	Range	Notes
usesStackingStrategy	Array_Architecture	Stacking_Strategy	

2.4 Causal properties

These are experimentally inferred, hold under conditions, and are multiply realised. Do **not** give them OWL characteristics such as functionality, and do not declare their causes disjoint — a real film has several at once.

Property	Domain	Range	Verified instances
stabilisesPhase	Material $\cup$ Defect $\cup$	Phase_State	TiN_capping stabilises Orthorhombic_Pca21 $S1\{I A\}$; Oxygen_Vacancy stabilises Orthorhombic_Pca21 and Tetragonal_P42nmc $S1\{I B, II A 2\}$; ZrO2_alloying stabilises Orthorhombic_Pca21 $S1\{I B\}$; TensileStrain stabilises Orthorhombic_Pca21 $S1\{II C\}$
destabilisesPhase	Material $\cup$ Defect $\cup$	Phase_State	Oxygen_Interstitial destabilises Orthorhombic_Pca21 by removing the barrier to monoclinic $S1\{II A 2\}$; ExcessOxygen stabilises Monoclinic_P21c $S1\{II A 2\}$
transformsPhase	Unit_Process	Phase_State	Domain is Unit_Process, not Thermal_Process. In Flow E an ALD electrode deposition performs the transformation $S24\{2\}$
scavengesOxygenFrom	Electrode_Material	Ferroelectric_Material	TiN, TaN scavenge; IrO2 does not, and degrades ferroelectricity $S1\{II A 2\}$; Ru has lower scavenging ability than TiN $S24\{4\}$
diffusesInto	Dopant $\cup$ Material	Material	Zr4 diffusesInto SiON, degrading the interfacial layer $S9\{p.5\}$
templates	Material	Material	$\Rightarrow$ UNCERTAIN t-phase grains sit within $< 5^\circ$ of both TiN layers, suggesting near-epitaxial growth during PMA. $S27$ says “may be some kind of” — implied, not demonstrated $S27\{p.1134\}$
exhibits	Memory_Cell_Architecture $\cup$ Layer_Stack	Degradation_Phenomenon	
mitigatedBy	Degradation_Phenomenon	Unit_Process $\cup$ Material	Fatigue mitigatedBy TiN relative to Ru $S24\{3.2\}$; Monoclinic_formation mitigatedBy Al2O3_insertion $S1\{II A 3\}$
enables	Capability	Array_Architecture $\cup$ Memory_Cell_Architecture	Conformality enables 3D_capacitor; LineOfSight blocks it $S1\{II B\}$
requiresCapability	Unit_Process $\cup$ Array_Architecture	Capability	
constrainsMaterialChoice	Capability	Material	$\Rightarrow$ UNCERTAIN Ru selected partly because dry plasma etch selective to HfO ₂ exists for it $S24\{1\}$ — stated as a motivation, not a demonstrated constraint

2.5 Measurement and evidence

Property	Domain	Range	Notes
hasMeasurement	Artifact $\cup$ Layer_Stack $\cup$ Memory_Cell_Architecture	Measurement	
measurementMethod	Measurement	Measurement_Method	
measuresProperty	Measurement	Measured_Property	
measuredOn	Measurement	Artifact	Distinguish blanket film / patterned capacitor / single cell / array. S12a measures both 1C and 1T-1C and reports they agree S12a{ $\llbracket 5$ }

3 Data properties

3.1 The Measurement class

Measurement is a reified n-ary relation. A bare `hasEnduranceCycles` on a material is misleading: S24's TiN device breaks down at 10^9 and its Ru device survives 10^{10} , at identical 3 μ s / 3 V pulses; S12a reports 10^9 at array level and 10^{11} at capacitor level from the same wafer.

```

Measurement
  measurementMethod      -> Measurement_Method
  measuresProperty        -> Measured_Property
  measuredOn              -> Artifact
  hasValue                xsd:double
  hasUnit                  xsd:string
  hasAppliedVoltage_V     xsd:double
  hasAppliedField_MVcm    xsd:double
  hasFrequency_Hz          xsd:double
  hasPulseWidth_s         xsd:double
  hasIntervalTime_s       xsd:double      S23{2}
  hasTemperature_C         xsd:double
  hasCycleCount            xsd:long
  hasBakeTime_min          xsd:double      S24{3.3}
  hasWaveform              { PUND, NDP, triangle, square, trapezoidal }
  hasVthConvention         { linear_extrapolation, constant_current }
                           S10{II A} vs S11{II}
  hasPolarisationStateType { SS, NSS, OS } S24{3.3}

```

3.2 Unit_Process

Property	Type	Verified range	Source
hasProcessTemperature_C	xsd:double	150–1050	S1, S9–S11
hasProcessDuration_s	xsd:double	1 to 14400	S10, S24
hasProcessPressure_mbar	xsd:double	7×10^{-4} to 2.7×10^{-1} (PLD); 1.2×10^{-2} to 5×10^{-2} (sputter)	S1 Tables III, IV
hasBasePressure_mbar	xsd:double	see S1 Table III	S1
hasGasFlow_sccm	xsd:double		S1{II B 2}
hasPulseTime_s	xsd:double	0.1, 0.5	S24{2}
hasPurgeTime_s	xsd:double	12	S24{2}
hasOxidantDoseTime_s	xsd:double	optimum 1–5	S1{II A 2}
hasCycleCount	xsd:int	65 supercycles for ~ 10 nm HZO	S24{2}
hasGrowthPerCycle_nm	xsd:double	< 0.05 for Cp precursors	S1{II A 1}
hasCoolingRate_Cps	xsd:double	4 and 10 compared	S1{I C}
hasImplantEnergy_keV	xsd:double	40, 50, 60	S10, S11
hasImplantDose_cm2	xsd:double	8×10^{12} , 2×10^{15} , 4×10^{15}	S10, S11
isCrystallising	xsd:boolean	true for E5, an ALD deposition	S24{2}

3.3 Material

Property	Type	Verified values	Source
hasFilmThickness_nm	xsd:double	0.7, 2, 9, 9.5, 10, 20 (FE); 3–100 (electrodes)	S9–S12a, S23, S24, S27
hasHfZrRatio	xsd:double	1:1 nominal; 0.8 measured	S9{p.2}, S24{2}
hasDopantConcentration_atPct	xsd:double	La 3.9, 4.5, 7.2; N 0.34 works, 2.44 kills FE	S27{p.1134}, S1 Table III
hasDopantConcentration_molPct	xsd:double	Y 1.9 (sputter), 3.8–5 (PLD), ~ 5 (CSD), 5.2 (CSD max Pr)	S1{II B 1, II D}
hasDopantRatio	xsd:string	Al:Hf = 1:30	S9{p.2}
hasBandgap_eV	xsd:double	> 5 ; 5.6	S1{I}, S23{1}
hasCarbonImpurity_atPct	xsd:double	TEMA 3.9, TDMA 2.3	S1{II A 1}
hasFilmDensity_gcm3	xsd:double	6.5–9.3 (TEMA); 8.5–12.8 (Cp)	S1{II A 1}
hasGrainLength_nm / hasGrainWidth_nm	xsd:double	HZO 12–30 $\times$ ~ 9 ; TiN bottom 6–15 equiaxed; TiN top 8–15 wide $\times$ ~ 30 tall	S27{p.1134}
hasCrystallisationOnset_C	xsd:double	HZO from 300; HAO above 650	S9{p.3}
hasTrapDensity_cm2	xsd:double	Et1: $1 \times 10^{14} \rightarrow 3.6 \times 10^{14}$; Et2: $\sim 1 \times 10^{14}$	S10{III D}

3.4 Phase_State and phase fraction

Property	Type	Notes
hasSpaceGroup	xsd:string	Pca2 ₁ , P4 ₂ /nmc, P2 ₁ /c, Fm-3m, R3m S1{I A}

Property	Type	Notes
isFerroelectric	xsd:boolean	true only for Pca2 ₁ (and $\Rightarrow$ UNCERTAIN contested R3m)
hasRelativeEnergy_meVperFU	xsd:double	o is +62 above m in pure HfO ₂ S1{I B}; Vo lower o by 15 and t by 24 S1{I B}
hasTransformationBarrier_meVperFU	xsd:double	t $\rightarrow$ m 223–263; t $\rightarrow$ o 22–31 S1{I C}
hasPermittivity_k	xsd:double	$\Rightarrow$ UNCERTAIN DISPUTED . S24 (citing Park 2013): m 16–20, o $\sim$ 30, t 35–40. S23 (citing Park 2016): o 25–30, t 35–40. Both values are secondary citations from the same research group’s work; neither S23 nor S24 measured phase permittivities directly. The t-phase value (35–40) is common to both
hasPhaseFraction	xsd:double	Must carry fractionMethod and phasesLumped
fractionMethod	xsd:string	{ GIXRD_intensity_ratio, PED_mapping, k_value_inference }
phasesLumped	xsd:string	{ “o+t”, “none” }

3.5 Memory_Cell_Architecture

Property	Type	Verified values	Source
hasCellArea_um2	xsd:double	0.009 (TSMC OS FeFET) $\Rightarrow$ UNCERTAIN trade press	S6
hasCapacitorDiameter_um	xsd:double	8, 10, 12, 16	S12a{¶4}
hasContactPadArea_cm2	xsd:double	$\approx 2 \times 10^{-5}$	S24{2}
hasGateLength_um / hasGateWidth_um	xsd:double	5/150; 5/150	S10, S11
hasArraySize_bit	xsd:int	16384 (16 kbit)	S12a
hasBitlineLength_bit	xsd:int	32, 64, 128, 256	S12a{¶6}
hasWriteVoltage_V	xsd:double	+4/ – 3.5 (MFIS), +5.5/ – 4 (MIFIS)	S11{III}
hasReferenceVoltage_V	xsd:double	1.8	S12a{¶6}

4 Verified process flows as ordered individuals

Flow	Steps in order	Anneal position	Source
A LabMFM	clean+BE $\rightarrow$ FE $\rightarrow$ TE $\rightarrow$ anneal 400–1000 °C / 1–60 s $\rightarrow$ patterned metal $\rightarrow$ SC1 etch	before patterning	S1{II, Fig. 3}
B1 GateLastFe-FET	implant $\rightarrow$ activation 1050 °C/5 s $\rightarrow$ dHF $\rightarrow$ IL (O ₃ , 300 °C) $\rightarrow$ HZO ALD 300 °C $\rightarrow$ TiN+W $\rightarrow$ anneal 550 °C/60 s $\rightarrow$ gate formation $\rightarrow$ contacts $\rightarrow$ FGA 450 °C	before patterning	S10{II A}
B2 GateLastFe-FET_MIFIS	B implant $\rightarrow$ S/D litho + As implant $\rightarrow$ activation 1050 °C/5 s $\rightarrow$ dHF $\rightarrow$ IL (O ₃ , 300 °C) $\rightarrow$ HZO (+Al ₂ O ₃) ALD 300 °C $\rightarrow$ TiN+W $\rightarrow$ anneal 400 °C/60 s $\rightarrow$ FGA 450 °C	before patterning	S11{II}
C BEOL_FeRAM	FEOL $\rightarrow$ M1–M6 $\rightarrow$ TaN BE $\rightarrow$ ALD HZO 20 nm $\rightarrow$ TaN TE $\rightarrow$ patterning and etch $\rightarrow$ anneal 500 °C $\rightarrow$ isolation $\rightarrow$ V6 via $\rightarrow$ M7	after patterning	S12a{Fig. 1a}
D MFIS_Capacitor	native oxide removal $\rightarrow$ chemical oxidation $\rightarrow$ NH ₃ RTP + O ₂ /N ₂ $\rightarrow$ ALD HfO ₂ 250 °C $\rightarrow$ TiN PVD $\rightarrow$ RTP 800 °C/20 s	before patterning (none described)	S9{p.2}

Flow	Steps in order	Anneal position	Source
E Ful- lALD_AutoAnneal	SiO ₂ /W → ALD TiN BE 400 °C → ALD HZO 240 °C → [REALD Ru] → ALD TiN TE 400 °C ≈4 h, crystallises → litho → plasma etch → Al pads	no anneal step	S24{2}
F MFM	pre-clean → ALD TiN BE 400 °C → ALD HZO 270 °C → ALD TiN TE + W → anneal 500 °C/60 s	before patterning (none described)	S23{2}
G LaHZO_MIM	Si → ALD TiN 10 nm → ALD La:HZO 9.5 nm → ALD TiN 30 nm → PMA 673 K, N₂, 1 h	before patterning (none described)	S27{p.1134}

Derived constraints:

- CrystallisationTemperature $\in [400, 800]$ °C across verified flows
- CrystallisationDuration $\in [20 \text{ s}, 4 \text{ h}]$ — three orders of magnitude
- Only Flow C inverts the patterning/anneal order
- Flows D, F, G describe no patterning step at all, which is a lack of evidence

5 Uncertainty register

5.1 Phase fraction is not directly measurable

The orthorhombic and tetragonal peaks overlap or sit very close in GI-XRD and SAED, making phase identification and quantification challenging. S27{p.1134, citing Chang et al., Acta Mater. 246 (2023)} The o111/t011 reflection sits near 30.3° 2θ. S1{I C}

Consequence: any hasPhaseFraction assertion needs fractionMethod and phasesLumped. The defensible practice is S24’s — quantify m only, lump o+t, then infer the split from the C-V permittivity S24{3.1}. But the permittivity calibration is itself disputed. S9 does report normalised o(111), o{200} and m(−111) intensities separately S9{p.3}, which is the practice S24 warns against.

Also: hasPhase on a film is a simplification. Films are polycrystalline mixtures; PED mapping resolves phase grain by grain S27{Fig. 1a}. Decide knowingly whether phase attaches to film, grain, or region.

5.2 Contested, conflicting, or implied

#	Item	Status
U1	Wake-up mechanism	S24 adopts Mehmood et al.’s two-step framework; its own contribution is showing T_{amb} dependence of $2P_r$ and leakage data that separates V_o redistribution (no leakage rise) from V_o generation (leakage rises at 10^5 cycles). S24’s chain: V_o redistribution and generation shift t/o relative stability → t→o transformation → wake-up S24{3.2}. S23 argues V_o migration reduces $E_{built-in}$ directly with no phase change, on the grounds that ε_r stays at ≈ 32.5 S23{3.2}. The ε_r datasets are not directly comparable : S23’s series spans 10^0 – 10^3 cycles; S24 locates the first wake-up stage between 10 and 10^5 – 10^6 cycles and measures k only at 10^5 and 10^8 . A third mechanism, ferroelastic switching, is untested and cannot be excluded S24{3.2}. A fourth, interfacial-layer degradation, is ruled out by S23 and judged unlikely by S24. Stack differences that may explain the divergence: oxidant (H ₂ O vs plasma O), T_{dep} (240 vs 270 °C), and crystallisation (400 °C/≈4 h vs 500 °C/60 s)

#	Item	Status
U2	Fatigue mechanism	Four candidates. Interfacial screening layer ruled out ; seed inhibition unlikely ; domain wall pinning and phase transformation both remain plausible S24{1, 3.2}
U3	Fatigue recovery by heating	Huang et al. report recovery by mild heating. S24 baked a fatigued Ru device at 105 °C for 1 s / 1 min / 1 h and found none — $2P_r$ fell further S24{3.2}
U4	Trapped charge trend during Fe-FET fatigue	S10 measures it increasing $17 \rightarrow 80 \mu\text{C}/\text{cm}^2$; Ichihara et al. (Kioxia) measured it decreasing . S10 attributes the difference to QSCV overestimating ΔQ_{Si} by 71.1% S10{I, II C}
U5	Curie temperature	Shiraishi $\sim 400\text{--}500^\circ\text{C}$; Hsain’s in-situ XRD shows the peak shift only from $\sim 800^\circ\text{C}$. Not reconciled; also thickness- and composition-dependent S1{I C}
U6	Seed layer direction	Park: a ZrO_2 starting layer degrades performance. Onaya: a bottom ZrO_2 layer gives <i>higher</i> P_r than no seed. S1 flags this as an open discrepancy S1{II A 3}
U7	“Antiferroelectric”	S1 insists on “antiferroelectric-like” because the origin is unproven S1{I}; S27 uses the bare term S27{p.1134}
U8	Permittivity calibration for o and t phases	Both values are secondary citations of Park-group work (S24 cites Park 2013; S23 cites Park 2016). Neither source measured phase permittivities directly. S24 gives o ~ 30 , t 35–40. S23 gives o 25–30, t 35–40. The t-phase figure (35–40) is common. The difference is a variation, not a fundamental conflict
U9	BEOL thermal ceiling	S24 asserts 400°C as the BEOL-compatible figure S24{1}; S12a and S23 both anneal at 500°C S12a{¶4}, S23{2}; S6 quotes $< 350^\circ\text{C}$ for oxide-semiconductor deposition. Not a fixed constraint
U10	Oxygen polarity rule is method-specific	Lower oxygen favours the ferroelectric phase in ALD and sputter; PLD is the opposite — more O_2 pressure gives more o-phase and higher P_r S1{II C}
U11	Phase pathway is method-specific	ALD/sputter: amorphous $\rightarrow$ t $\rightarrow$ o on cooling, irreversible $\rightarrow$ m on further heating. PLD/epitaxial: m $\rightarrow$ t $\rightarrow$ o on heating S1{II C}
U12	Thickness ceiling is method-specific	ALD/sputter tens of nm; PLD to 930 nm; CSD to $1 \mu\text{m}$ S1{II C, II D}
U13	La doping does not guarantee the o-phase	At 7.2 at.% La with a 400°C / 1 h PMA, PED shows the majority phases are tetragonal and monoclinic S27{p.1134}. The in-situ biasing experiment stated in S27 was “in progress” at time of publication — no results are available
U14	Electrode substitution is multi-objective	Ru vs TiN: endurance improves ($10^9 \rightarrow \geq 10^{10}$) while fatigue becomes severe ($2P_r \rightarrow 7 \mu\text{C}/\text{cm}^2$ at 10^{10}), m-phase rises $22 \rightarrow 38\%$, OS retention improves slightly, SS retention worsens. No dominant option S24{3.2, 3.3, 4}
U15	Imprint’s effect on P_r is waveform-dependent	Visible under NDPU, absent under PUND because PUND saturates the polarisation above 3 MV/cm S23{3.1, 4}
U16	V_{th} conventions differ	Linear extrapolation S10{II A} vs constant current S11{II}, both on a B1500A. Memory windows are not directly comparable
U17	Templating by the electrode	t-phase grains within $< 5^\circ$ of both TiN layers suggest near-epitaxial growth during PMA — S27 says “may be some kind of” S27{p.1134}
U18	Wake-up as a production step	Verified only as an experimental preparation S24{3.2}. v1’s claim that it is a manufacturing step rested on an unobtainable source
U19	FTJ	Named as an application S1{I}, S27{p.1134}. No structure or data verified
U20	S12a’s headline device numbers	30 ns switching, $> 10^4$ s retention, $> 10^{11}$ capacitor cycles appear in the Conclusions with supporting data in Appendices A–D, which are not available. Only the array figure ($> 10^9$ at 3.5 V, $1 \mu\text{s}$) has verified conditions

#	Item	Status
U21	Internal inconsistencies in verified sources	S1{II A 2} ozone/water P_r figures and ozone dose figures conflict with S1 Table I; S1 Fig. 3 caption conflicts with the body on step 5; S11 abstract says 10^5 cycles, body says 7×10^4 ; S23 abstract says 10 s max interval, Fig. 2 caption says 30 s.

5.3 Absences to preserve

Do not let a closed-world reading turn these into negative facts:

- Flows D, F, G describe no patterning step. They were capacitor studies; patterning may simply be undescribed.
- No verified source describes packaging, wafer-level burn-in, or ellipsometry/XRR/AFM/SEM metrology. v1 asserted these; they are removed rather than negated.
- No verified source names a lithography exposure tool.

6 Sources

All ten were read in full or near-full. Nothing in this specification derives from a search snippet, an abstract, or a source not listed here.

Tag	Citation	Extent
S1	Hsain, Lee, Materano, Mittmann, Payne, Mikolajick, Schroeder, Parsons, Jones. <i>Many routes to ferroelectric HfO₂: A review of current deposition methods</i> . J. Vac. Sci. Technol. A 40 (1), 010803 (2022). doi 10.1116/6.0001317	Full text: I, I A–C, II, II A 1–3, II B 1–3, II C, II D, II E, III; Tables I–V; Figs. 1–21
S5	Duan, Khan, Gong, Narayanan, Ni. <i>A Full Spectrum of 3D Ferroelectric Memory Architectures Shaped by Polarization Sensing</i> . arXiv:2504.09713 (2025)	Abstract → “Parallel Stacking 3D Integration”
S6	LaPedus, M. <i>Next-Gen Ferroelectric Memory: Still A Work In Progress</i> . Semiecosystem, 8 Feb 2026	Full text. Reports IEDM Dec 2025 papers not read directly
S9	Yang, Lehninger, Reinig, Schöne, Hoffmann, Seidel, Lederer, Gerlach. <i>Enhanced Performance of FeFET Gate Stack via Heterogeneously co-doped Ferroelectric HfO₂ Films</i> . arXiv:2508.16768 (2025). Fraunhofer IPMS CNT / TU Dresden	Full text, 6 pp. MFIS capacitors, not FeFETs
S10	Zhao, Tian, Xu, Xiang, Li, Chai, Duan, Han, Wang, Wang, Ye. <i>Experimental Extraction and Simulation of Charge Trapping during Endurance of FeFET with TiN/HfZrO/SiO₂/Si (MFIS) Gate Structure</i> . arXiv:2106.15939 (2021). IME, Chinese Academy of Sciences	Full text
S11	Hu, Sun, Bai, Jia, Dai, Li, Han, Ding, Fan, Zhao, Chai, Xu, Si, Wang, Wang. <i>Enlargement of Memory Window of Si Channel FeFET by Inserting Al₂O₃ Interlayer on Ferroelectric Hf_{0.5}Zr_{0.5}O₂</i> . arXiv:2312.16829 (2023). IME CAS / SJTU	Full text, 4 pp.
S12a	Xiao, Peng, Liu, Duan, Bai, Yu, Ren, Yu, Han, Hao. <i>Hf_{0.5}Zr_{0.5}O₂ 1T–1C FeRAM arrays with excellent endurance performance for embedded memory</i> . Sci. China Inf. Sci. (2022). doi 10.1007/s11432-022-3469-5	Full 2-page Letter. Appendices A–D unavailable

Tag	Citation	Extent
S23	Ding, Weng, Lan, Yan, Cai, Qu, Zhao. <i>Wake-Up and Imprint Effects in Hafnium Oxide-Based Ferroelectric Capacitors during Cycling with Different Interval Times</i> . Electronics 13 , 1021 (2024). doi 10.3390/electronics13061021	Full text
S24	Khakimov, Chernikova, Koroleva, Markeev. <i>On the Reliability of HZO-Based Ferroelectric Capacitors: The Cases of Ru and TiN Electrodes</i> . Nanomaterials 12 , 3059 (2022). doi 10.3390/nano12173059. MIPT	Full text, 16 pp.
S27	Parida, Favia, Richard, De, Popovici, Grieten. <i>Study of Ferroelectric Phases in HfZrO_x Thin Films with Varying La Content Using Precession Electron Diffraction & In-Situ Heating-Biasing Techniques</i> . Microsc. Microanal. 31 (7), 1134–1135 (2025). doi 10.1093/mam/ozaf048.588. IMEC	Full 2-page abstract. Key in-situ biasing experiment stated as “in progress”